Open Beauty Facts collection API

`src.collection` provides two deliberately separate collection modes. It never creates risk labels, including for records whose label would be unknown in a 16-item registry.

The default source is **Open Beauty Facts**, not Open Food Facts:

```
Plain Text

https://world.openbeautyfacts.org/api/v2/product/{barcode}.json
```

Small live API sample

A live sample is expressly limited to one to three product endpoint calls. Requests are serial, send a custom `User-Agent`, use a 5-second connect / 30-second read timeout by default, and retry network failures, HTTP 429, and 500/502/503/504 with bounded exponential backoff plus jitter. This aligns with the published 15 product-read requests per minute per IP limit. The caller should replace the default User-Agent with an application name and reachable contact before production use. [2]

From the project root, reproduce the executed sample with:

```
Bash

python -m src.collection \
  --from-labeled-csv /home/ubuntu/upload/products_labeled.csv \
  --limit 3 \
  --output data/api/product_snapshot.json
```

For this mode, the collector deterministically prefers the first records with a code of at least four characters, a non-empty product name, and non-empty ingredient text. It uses another valid code only if it cannot fill the requested sample. This avoids selecting very short legacy identifiers when stronger known product examples are already present in the supplied CSV; it does not pad, cast, or otherwise alter any selected barcode text.

The output is an auditable JSON snapshot. Every result records the barcode **as text**, request URL, timestamp, number of attempts, HTTP status, outcome, and actual decoded JSON response. An unavailable endpoint, a non-2xx response, or a 2xx response whose Open Beauty Facts payload does not report `status: 1` is retained as a diagnostic; the collector does not manufacture a product, ingredient list, label, or risk result.

The `provided_labeled_csv` metadata intentionally preserves this existing dataset provenance:

- `original_download_date`: `2026-10-10`

- `original_source` : `fr.openbeautyfacts.org export`
- scope note: an export hosted at `fr.openbeautyfacts.org` is **not** proof that every product is France-only.

That metadata describes the original labeled CSV. It does not make the fresh API response historical or assert a country of sale for an individual product.

For explicit codes rather than a CSV, use:

```
Bash

python -m src.collection --barcodes 00001775 00001785 --output data/api/product
```

No network request is made without one of `--barcodes` , `--from-labeled-csv` , or `--bulk-csv` .

Optional bulk cosmetics CSV

The bulk option is intentionally not a live API default and must receive a destination path:

```
Bash

python -m src.collection --bulk-csv data/raw/openbeautyfacts-products.csv.gz
```

Its default source is the official cosmetics export:

```
Plain Text

https://static.openbeautyfacts.org/data/en.openbeautyfacts.org.products.csv.gz
```

The bulk file is an offline, historical export generated nightly, suited to whole-dataset analysis or caching. It is not the newly executed API sample. On completion, the command writes a sibling `.metadata.json` file containing source URL, retrieval time, response status, selected response headers, byte count, and SHA-256. The official data page specifies UTF-8 and tab-separated fields for the CSV export. [1]

Use the export rather than turning the product endpoint into a crawl: the official API guidance asks users who need more than a few hundred products to download CSV or JSONL data directly. [2]

Verified execution

The committed `data/api/product_snapshot.json` was newly executed at `2026-10-31T02:38:53Z` by the API-sample command above, not by the bulk export. The verified requests were:

- `https://world.openbeautyfacts.org/api/v2/product/1785.json` — HTTP `200`, API `status: 1`
- `https://world.openbeautyfacts.org/api/v2/product/1959.json` — HTTP `200`, API `status: 1`
- `https://world.openbeautyfacts.org/api/v2/product/2337.json` — HTTP `200`, API `status: 1`

All three saved responses contain a product object with non-empty `ingredients_text`. The top-level `metadata.source_base_url`, per-result `request_url`, and per-result `status_code` remain the source of truth for the exact execution. The snapshot contains no synthetic fallback data.

Data reuse

Open Beauty Facts publishes the database under ODbL, with separate terms for contents and images. Applications that publicly reuse the data need appropriate attribution and should review the official terms. `1`

References

[1] Open Beauty Facts data, exports and product API
[2] Product Opener API usage and rate-limit documentation